

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and deployed **predictive analytics models** using **Python (pandas, scikit-learn, NumPy)** to forecast payment transaction volumes and anomaly patterns across banking systems, achieving **92% accuracy** in detection of high-risk SWIFT messaging flows and reducing incident response time by **35%**.
- Built end-to-end **data pipelines** and **ETL workflows** using **Python** and **SQL (PostgreSQL, Oracle, MySQL)** to ingest, validate, and transform multi-source banking transaction data; automated daily reporting across **10+ enterprise systems** with **zero data loss** and **99.8% uptime**.
- Designed and published **interactive BI dashboards** in **Power BI** and **Tableau** for banking operations and finance stakeholders, visualizing ISO 20022 message flows, SEPA payment reconciliation, and SLA compliance metrics; reduced manual reporting overhead by **40 hours/month**.
- Performed **statistical analysis** and **hypothesis testing** on payments-processing data using **R** and **Python** to identify root causes of production incidents and validate optimization hypotheses; delivered insights that improved transaction throughput by **18%** and reduced error rates to **0.2%**.
- Automated **data quality validation** and **monitoring workflows** using **Python scripting, SQL, and AWS Lambda** to flag schema violations, missing reconciliation records, and compliance breaches in real-time; prevented **12 regulatory audit findings** through proactive data governance.
- Integrated **Claude API (Anthropic)** and **prompt engineering** into a production AI automation pipeline for tailored resume generation at scale; orchestrated **end-to-end data workflows** from web scraping through structured-JSON transformation and PDF export, demonstrating proficiency in **LLM-driven data processing** and **AI-powered analytics automation**.

SKILLS

Data Science & Machine Learning – Predictive modelling, supervised learning, unsupervised learning, regression analysis, classification, anomaly detection, model evaluation metrics, scikit-learn, evaluation frameworks.

Data Engineering & ETL – Python (pandas, NumPy), SQL, ETL pipeline development, data transformation, data validation, PostgreSQL, MySQL, Oracle, data warehousing, scheduling.

Business Intelligence & Visualization – Power BI, Tableau, DAX, Power Query, data modelling, interactive dashboards, stakeholder reporting.

Statistics & Analysis – Exploratory data analysis, hypothesis testing, statistical methods, time-series analysis, A/B testing, R, RStudio.

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux, Git, Jenkins CI/CD pipelines.

Domain & Compliance – Banking & financial services, payments processing, SWIFT messaging (MT/MX), ISO 20022 message flows, SEPA, cross-border payments, reconciliation workflows, regulatory compliance, SLA management.

AI & Automation – Claude API, Anthropic SDK, prompt engineering, LLM-driven workflows, API integration, structured-JSON processing, GitHub Actions, Vercel deployment.

EDUCATION AND TRAINING

PROJECTS

SWIFT Payments Analytics & Compliance Dashboard (HCL Technologies)

- Built **multi-dimensional Power BI dashboard** ingesting ISO 20022 message flows, SEPA transaction data, and cross-border payment reconciliation records from PostgreSQL and Oracle data warehouses; visualized **real-time SLA compliance**, transaction-flow anomalies, and regulatory audit trails for banking operations stakeholders.
- Developed **Python-based ETL pipeline** to extract SWIFT MT/MX messaging metadata, validate message formats, and load normalized transaction data into a centralized warehouse; achieved **99.8% data accuracy** and reduced manual reconciliation time by **30 hours/week**.
- Implemented **anomaly detection model** using **scikit-learn isolation forests** on payment transaction features; flagged high-risk flows and unusual patterns with **93% precision**, enabling faster incident response and regulatory compliance verification.

Ireland Jobs Tracker – LLM-Powered Resume Automation (Personal Project)

- Architected and deployed **end-to-end AI data pipeline** leveraging **Claude API (Anthropic)** and **prompt engineering** to autonomously generate job-description-tailored resumes at scale; integrated **Apify web scrapers**, structured-JSON payload transformation, and **reportlab PDF generation**.
- Engineered **multi-stage data workflows** combining job-description ingestion, constraint-based prompt engineering, and LLM-driven content synthesis; orchestrated orchestration via **GitHub Actions CI/CD** and **Vercel deployment**, demonstrating production-grade **AI-driven analytics automation**.
- Optimized **LLM prompt constraints** and response validation to ensure consistency and compliance with formatting rules; processed **100+ unique job descriptions** with **zero manual post-processing**, validating the scalability of LLM-powered data transformation pipelines.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)